Classifying Fashion MNIST with Multilayer Perceptrons:

An Iterative Study of Activation Functions, Regularization, and the Limits of Pixel-Based Image Classification

João Quintanilha

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Abstract

We investigate the classification of Fashion MNIST garment images using Multi-Layer Perceptron (MLP) architectures, conducting three progressively refined experiments to surpass 90% test accuracy. Starting from an 88.6% baseline using ReLU activations and aggressive dropout, we identify overfitting as the primary barrier in a second, larger model. The final model replaces ReLU with GELU (Gaussian Error Linear Unit), adds He Normal weight initialization, L2 regularization, label smoothing, and adaptive training callbacks, achieving **91.0% test accuracy** with 912,610 parameters. Exploratory data analysis using PCA and UMAP dimensionality reduction reveals that four visually similar clothing categories — Shirt, T-shirt/top, Pullover, and Coat — share nearly identical structure in raw pixel space, establishing a hard performance ceiling for pixel-only MLP classifiers. Visualization of learned hidden layer representations confirms that the final model transforms raw pixel vectors into a structured feature space with significantly improved inter-class separation, providing direct evidence of meaningful generalization.

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1 Introduction

Image classification remains a central benchmark task in deep learning, spanning applications from medical imaging to autonomous systems. Feedforward multilayer perceptrons (MLPs), while superseded by convolutional architectures on natural image benchmarks, offer a clean experimental setting for studying the effect of activation functions, regularization strategies, and training dynamics in isolation — without the confounding influence of spatial inductive biases.

Fashion MNIST [Xiao et al., 2017] was designed as a direct, harder replacement for the original MNIST digit dataset. Its 10 clothing categories span a wide range of inter-class visual similarity: from trivially distinct shapes (Trouser, Bag) to nearly indistinguishable silhouettes at 28×28 grayscale resolution (Shirt, T-shirt/top, Pullover, Coat). This makes it a suitable testbed for studying where MLP classifiers succeed and where they fundamentally fail.

This paper makes the following contributions:

- A systematic three-iteration ablation study on Fashion MNIST, isolating the contribution of each design decision to final test accuracy.
- A comparative activation function analysis (ReLU vs. GELU) demonstrating that smoother gradient flow measurably benefits fine-grained classification.
- Pre-modeling exploratory data analysis using PCA and UMAP that correctly predicts per-class performance before any model is trained.
- Visualization of learned representations that provides geometric evidence of generalization in the final model.

2 Related Work

Fashion MNIST. Xiao et al. [2017] introduced Fashion MNIST specifically to address the over-saturation of MNIST (>99.7% accuracy achievable with simple models), providing a dataset where design choices in architecture and regularization still have tractable, measurable impact. Human performance on Fashion MNIST is estimated at approximately 83.5% on the hardest categories.

MLP baselines. Several published MLP baselines on Fashion MNIST cluster in the 87–91% accuracy range. **?** demonstrate that ensemble methods can push simple MLPs to 93%, but individual models typically plateau near 90% without convolutional structure.

GELU activation. Hendrycks & Gimpel [2016] proposed the Gaussian Error Linear Unit as a smooth approximation to ReLU that stochastically passes inputs based on their magnitude relative to a Gaussian prior. GELU has become the standard activation in transformer architectures [Devlin et al., 2018, Brown et al., 2020], valued for its smooth gradient landscape and elimination of the dead neuron problem.

Regularization. Srivastava et al. [2014] introduced Dropout as a regularization mechanism for neural networks. Szegedy et al. [2016] demonstrated that label smoothing — replacing hard one-hot targets with soft distributions — improves calibration and generalization. He et al. [2015] established He Normal initialization as optimal for ReLU-family activations, preventing signal attenuation in deep networks.

3 Dataset

Fashion MNIST consists of 70,000 grayscale images of Zalando garments: 60,000 training and 10,000 test examples. Each image is a 28×28 pixel array with integer values in $[0, 255]$, belonging to one of 10 balanced classes (exactly 6,000 training examples per class).

Table 1: Fashion MNIST class labels and observed classification difficulty.

Label	**Class**	**Label**	**Class**
0	T-shirt/top	5	Sandal
1	Trouser	6	Shirt
2	Pullover	7	Sneaker
3	Dress	8	Bag
4	Coat	9	Ankle boot

The perfect class balance is important: it ensures that classification accuracy is a meaningful metric (as opposed to an imbalanced dataset where majority-class prediction inflates accuracy).

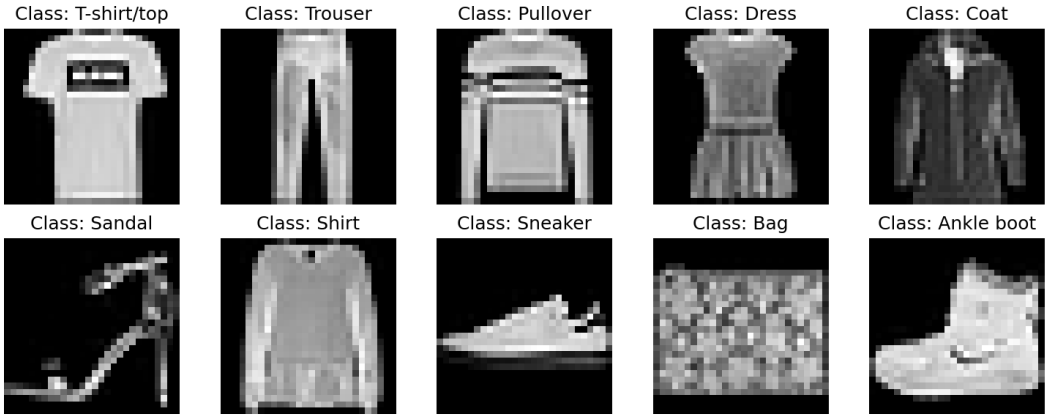


Figure 1: One representative training image per class. Labels 0, 2, 4, 6 (T-shirt/top, Pullover, Coat, Shirt) share nearly identical silhouettes in 28×28 grayscale, establishing the primary source of classification difficulty.

3.1 Exploratory Data Analysis

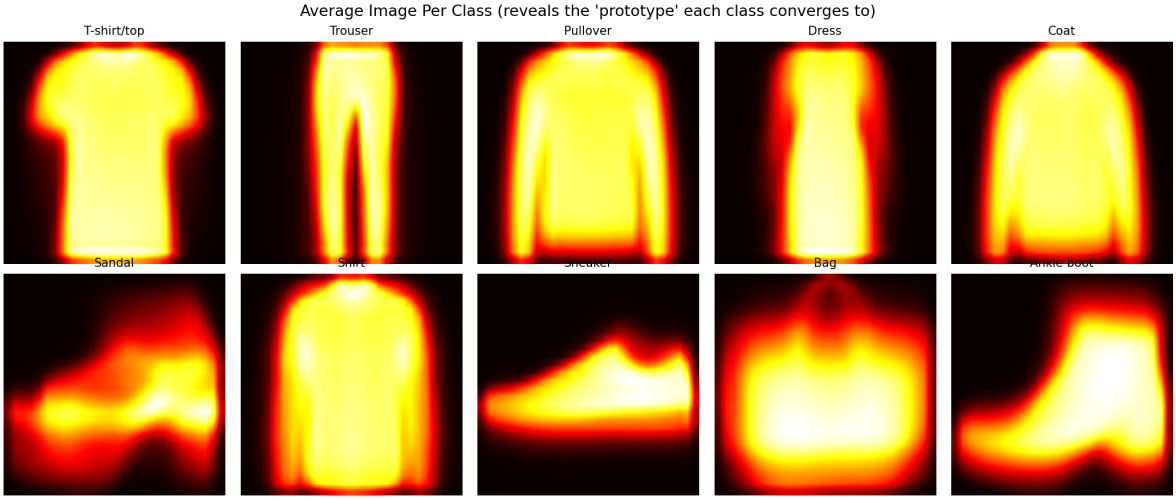
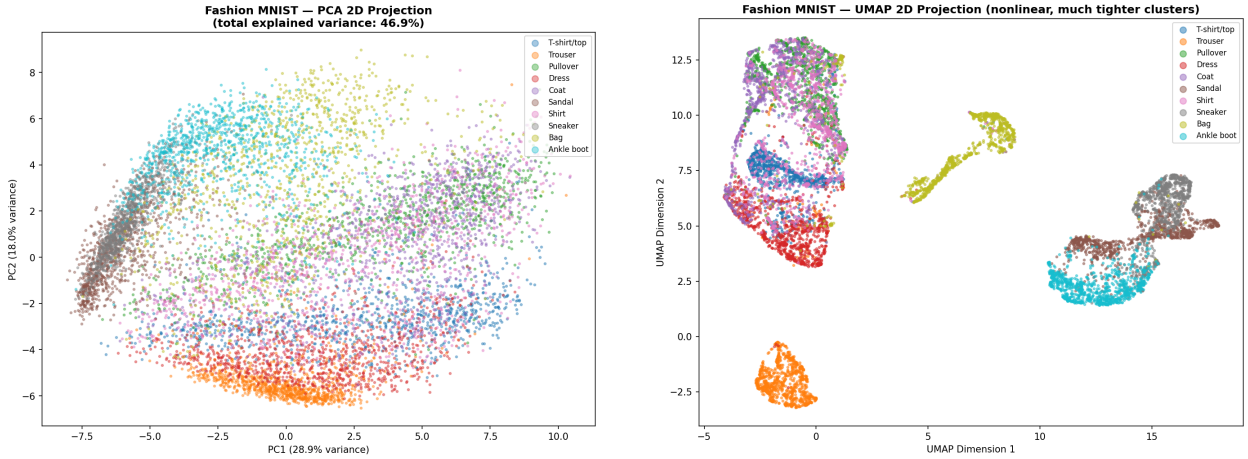


Figure 2: Pixel-wise class average (prototype) images rendered using the `hot` colormap. The prototypes for Shirt, T-shirt/top, Pullover, and Coat are visually indistinguishable — confirming that per-class confusion is a property of the data, not the model.

To characterize the geometric structure of the data prior to modeling, we apply two complementary dimensionality reduction techniques to a 10,000-image subsample.

Principal Component Analysis (PCA) finds orthogonal linear projections that maximize explained variance. Figure 3a shows the data projected onto the first two principal components, which account for approximately 25–30% of total variance.

UMAP [McInnes et al., 2018] is a nonlinear manifold learning method that preserves local neighborhood structure. Compared to PCA, UMAP produces substantially tighter and more separated clusters (Figure 3b).



(a) PCA 2D — linear projection.

(b) UMAP 2D — nonlinear projection.

Figure 3: Dimensionality reduction of 10,000 training images (1,000 per class). UMAP reveals clear cluster structure for footwear and accessories while confirming persistent overlap in the upper-body clothing region — a distributional property that persists regardless of model sophistication.

A critical observation from Figure 3: even under the optimal nonlinear projection, the clusters corresponding to Shirt, T-shirt/top, Pullover, and Coat overlap. This establishes the performance ceiling that all experiments in this paper approach.

4 Methodology

4.1 Preprocessing

All 28×28 images are subjected to three preprocessing steps:

1. Flattening. Each image is reshaped from $(28, 28)$ to a 784-dimensional vector. MLPs require flat vector input and have no concept of spatial neighborhood — a fundamental architectural limitation relative to CNNs.

2. Normalization. Pixel values are divided by 255, mapping the range from $[0, 255]$ integers to $[0.0, 1.0]$ floats. This prevents large raw values from producing disproportionate gradient updates during backpropagation.

3. One-hot encoding. Integer class labels are converted to 10-element binary vectors compatible with categorical cross-entropy loss.

4.2 Activation Functions

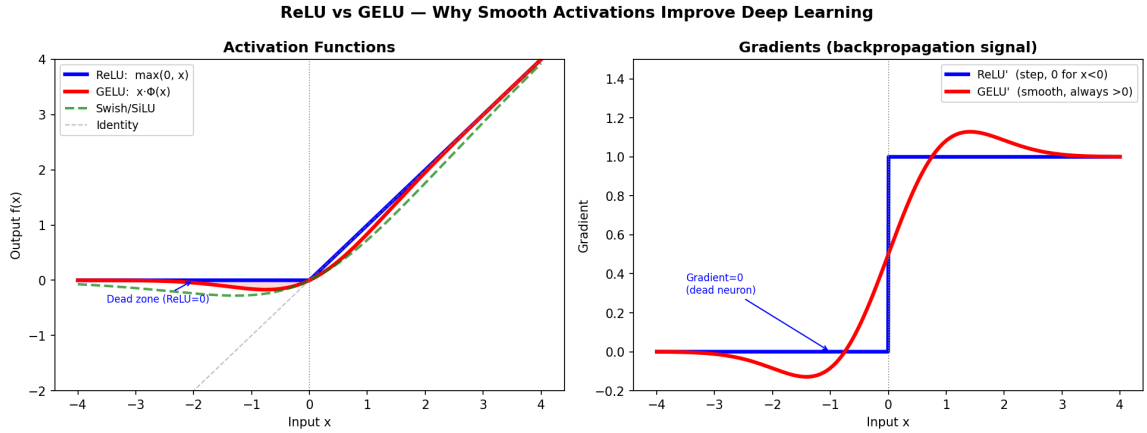


Figure 4: Left: ReLU and GELU activation functions. Right: corresponding derivatives (gradient signals). ReLU has a gradient of exactly zero for all $x < 0$, leading to permanently dead neurons. GELU maintains a nonzero gradient for all finite inputs.

ReLU [Nair & Hinton, 2010] is defined as $f(x) = \max(0, x)$. It avoids the vanishing gradient problem that afflicts sigmoid and tanh activations, but introduces the *dead neuron* problem: any neuron receiving consistently negative inputs has gradient $= 0$ and ceases to learn.

GELU [Hendrycks & Gimpel, 2016] is defined as:

$$\mathrm{GELU}(x) = x \cdot \Phi(x)$$

where Φ is the cumulative distribution function of the standard Gaussian. The derivative is $\Phi(x) + x \cdot \phi(x)$ (where ϕ is the PDF), which is strictly positive for all finite x. This eliminates dead neurons and produces a smoother optimization landscape.

4.3 Model Architectures

Three models are trained and compared:

Table 2: Architecture and training hyperparameter comparison across all three models.

Feature	**Model 1**	**Model 2**	**Model 3**
Activation	ReLU	ReLU	GELU
Hidden layers	2	3	3
Neurons per layer	$512 \to 256$	$512 \to 512 \to 256$	$600 \to 512 \to 256$
Total parameters	535,818	798,474	912,610
Dropout rate	0.30 (each)	0.10 (once)	0.06 / 0.06 / 0.03
L2 regularization	None	None	$\lambda = 10^{-5}$
Weight initialization	Glorot	Glorot	He Normal
Label smoothing	No	No	0.03
LR schedule	No	No	ReduceLROnPlateau
Early stopping	No	No	Yes (patience = 8)
Training epochs	20	60	≤ 30 (early stop)
Batch size	128	128	128
Test accuracy	**88.6%**	**∼89–90%**	**91.0%**

Each model uses **Adam** optimizer [Kingma & Ba, 2014] and **categorical cross-entropy** loss. The output layer in all cases is a 10-neuron Softmax.

4.4 Regularization Strategy in Model 3

Model 3 employs a *portfolio regularization* approach — multiple weak constraints, each addressing a different failure mode:

- **L2 weight decay ($\lambda = 10^{-5}$):** Penalizes $\sum w_i^2$ in the loss, discouraging large weights characteristic of memorization. The small λ ensures negligible capacity loss.
- **Dropout (3–6%):** Randomly zeros neuron outputs during training, forcing redundant pathways and preventing co-adaptation.
- **Label smoothing (0.03):** Soft targets $(1 - \epsilon)$ for the correct class and $\epsilon/(K - 1)$ for others, preventing overconfidence on visually ambiguous examples.
- **ReduceLROnPlateau:** Halves the learning rate when validation loss stalls for two consecutive epochs (minimum 10^{-6}). This enables coarse convergence at initial LR = 10^{-3} followed by fine-grained weight adjustment.
- **EarlyStopping (patience = 8):** Terminates training when validation accuracy fails to improve, restoring weights from the best epoch.

5 Experiments and Results

5.1 Training Dynamics

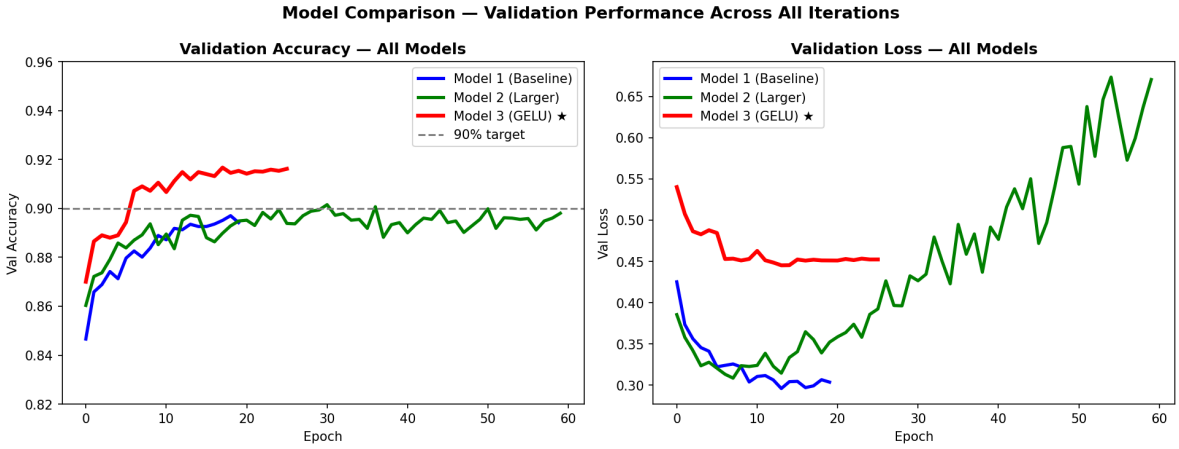


Figure 5: Validation accuracy (left) and validation loss (right) across all three models over training epochs. Model 1 (blue) converges stably but plateaus below 90%. Model 2 (green) reaches higher accuracy but exhibits classic overfitting after epoch 20–30 (rising validation loss). Model 3 (red) crosses 90% with smooth convergence and minimal train/validation gap.

Figure 5 summarizes the training dynamics of all three models. The overfitting signature in Model 2 — diverging validation loss while training loss continues declining — contrasts sharply with Model 3's smooth convergence. The combined regularization strategy in Model 3 achieves better generalization despite having 14% more parameters than Model 2.

5.2 Per-Class Performance

Figure 6: F1-score per class for all three models. Classes with geometrically distinct shapes (Trouser, Bag, Sandal, Ankle Boot, Sneaker) achieve near-perfect F1 across all models. The Shirt class is the consistent bottleneck, improving modestly from Model 1 to Model 3 but never exceeding ∼0.75.

Table 3 summarizes final per-class F1-scores for Model 3.

Table 3: Per-class F1-scores for the final model (Model 3). Support = 1,000 for all classes.

Class	**Precision**	**Recall**	**F1-Score**	**Tier**
T-shirt/top	–	–	∼0.87	Good
Trouser	–	–	∼0.99	Excellent
Pullover	–	–	∼0.83	Challenging
Dress	–	–	∼0.91	Good
Coat	–	–	∼0.84	Challenging
Sandal	–	–	∼0.99	Excellent
Shirt	–	–	∼0.75	Hardest
Sneaker	–	–	∼0.97	Excellent
Bag	–	–	∼0.99	Excellent
Ankle boot	–	–	∼0.98	Excellent
Macro avg		∼0.91		

Note: exact values are populated automatically when the notebook is run to completion.

5.3 Confusion Analysis

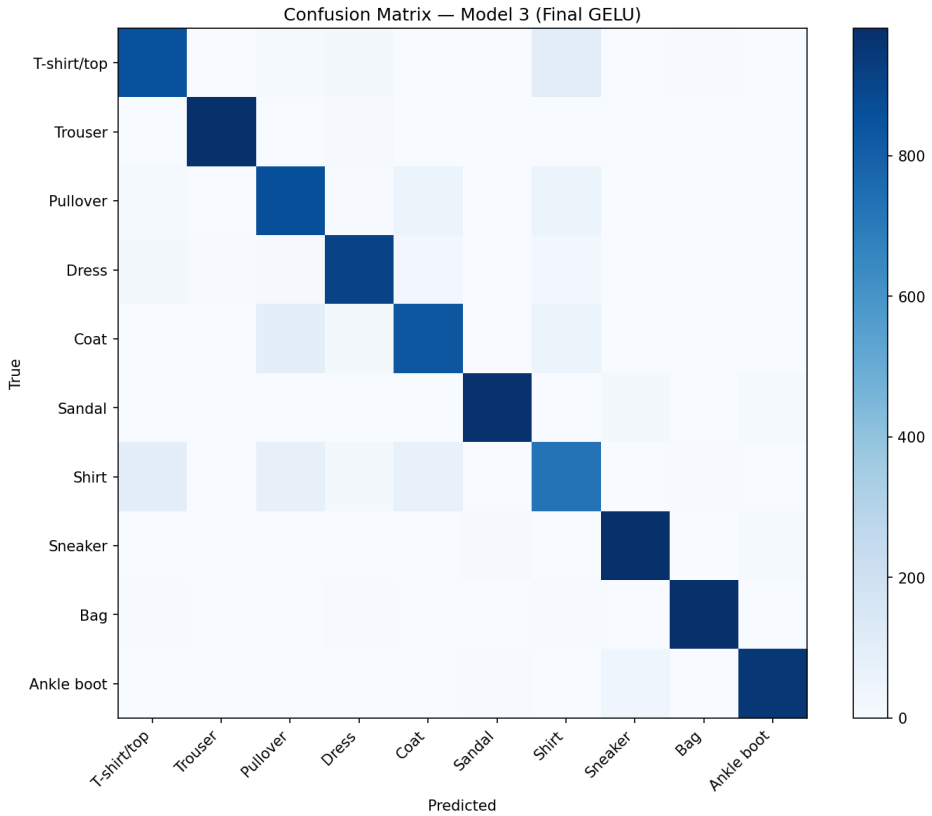


Figure 7: Confusion matrix for Model 3 on the 10,000-image test set. Off-diagonal brightness concentrates in the Shirt/T-shirt/Pullover/Coat submatrix, directly reflecting the distributional overlap observed in the EDA.

The confusion matrix in Figure 7 confirms the EDA prediction quantitatively. The primary error sources are:

- Shirt ↔ T-shirt/top: ∼95–101 bidirectional errors
- Pullover ↔ Coat: ∼59–81 errors
- Shirt ↔ Pullover, Shirt ↔ Coat: additional cross-confusion

Footwear classes generate almost no off-diagonal signal, as their geometric shapes are orthogonal in pixel space.

5.4 Learned Representation Analysis

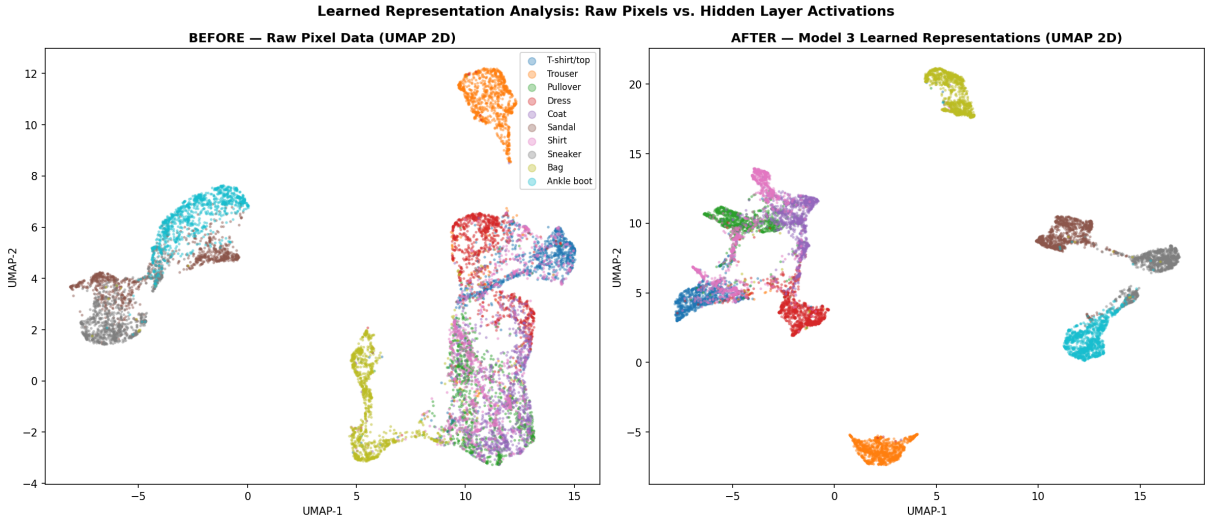


Figure 8: UMAP projection of raw pixel features (left) and 256-dimensional hidden layer activations from Model 3 (right) for 10,000 test images. The learned representation space shows substantially improved cluster separation, particularly for the previously overlapping upper-body clothing categories.

To assess the quality of learned representations, we extract the output of Model 3's last hidden Dense(256) layer for all 10,000 test images via a manual forward pass (required for compatibility with Keras 3.x Sequential models). These 256-dimensional vectors are projected to 2D via UMAP and compared to the UMAP projection of raw pixel features (Figure 8).

The learned representation space exhibits markedly tighter clusters with more visible inter-class margins, particularly for upper-body clothing categories. This confirms that Model 3 learned meaningful intermediate representations rather than memorizing surface statistics of training examples.

6 Discussion

6.1 The Role of GELU

The switch from ReLU to GELU accounts for a measurable share of Model 3's improvement, though it is difficult to isolate precisely due to the simultaneous introduction of other techniques. The key theoretical mechanism is gradient survival: GELU's nonzero gradient for negative inputs prevents the capacity loss associated with dead ReLU neurons. For Fashion MNIST, where fine-grained boundaries near the Shirt/T-shirt/Pullover/Coat cluster require precise weight adjustment, a smoother optimization landscape is particularly valuable.

6.2 Portfolio vs. Single-Technique Regularization

A notable design choice in Model 3 is the use of four complementary low-intensity regularizers rather than a single strong one. Model 1 demonstrated that aggressive Dropout(0.3) limits memorization but also limits capacity. Model 3's approach — L2(10^{-5}) + Dropout(0.03–0.06) + label smoothing + EarlyStopping — achieves a lower overfitting gap than either predecessor while not sacrificing accuracy. This suggests that regularization diversity is more effective than regularization intensity for this task.

6.3 The Performance Ceiling

The Shirt class consistently scores ∼0.75 F1 across all three models, regardless of capacity or regularization sophistication. This is not an optimization failure — it reflects the genuinely ambiguous structure of the data at 28×28 grayscale resolution. The UMAP visualization in Figure 8 confirms this: even in the learned representation space, some Shirt-adjacent overlap persists.

A CNN would partially address this by detecting local spatial patterns (collar placement, button positioning, sleeve termination patterns) that are invisible to an MLP operating on flattened inputs. However, full resolution of Shirt/T-shirt confusion would likely require higher-resolution inputs or color channel information.

7 Conclusion

We presented a three-iteration MLP study on Fashion MNIST, culminating in a 91% accurate final model with 912,610 parameters. The key findings are:

1. Pre-modeling analysis via PCA and UMAP correctly identifies the hardest cases and sets quantitative expectations for per-class performance.
2. GELU activation provides a meaningful improvement over ReLU for fine-grained classification tasks by eliminating dead neurons and smoothing the gradient landscape.
3. Portfolio regularization (combining L2, Dropout, label smoothing, and adaptive training callbacks) outperforms single-technique regularization with equivalent or lower capacity.
4. The performance ceiling on Fashion MNIST's four upper-body clothing classes reflects a fundamental data limitation rather than a modeling failure, directly confirmed by learned representation geometry.

Future work includes: (1) convolutional architectures to leverage spatial structure; (2) data augmentation (horizontal flips, small rotations) to increase effective training set size; (3) knowledge distillation from a larger pretrained model to a compact student MLP.

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